CO2xN: Porewater Sulfide

July 2026 Samples

2026-07-23

Code Set up

run information

```
##Before or After Reruns (YES or NO)
After_Reruns = "YES"
#If NO will not write final data file
  #And Will Write a Rerun File

###things that need to be changed

sample_month = "July"
sample_year = "2026"
Run_Date = as.Date("07/23/2026", format = "%m/%d/%Y")
Run_by = "Melanie Giessner"#Instrument user
Script_run_by ="Melanie"#Code user

Run_notes=""#any notes from run

#Import Plate Data

folder_path_rawdata <- paste0("Raw Data")
Raw_plates <- list.files(path = folder_path_rawdata, full.names = TRUE,pattern = ("(202607|202608)(.+Plate
print(Raw_plates)
```

```
##  [1] "Raw Data/20260716_LTREB_H2S_Plate1.csv"
##  [2] "Raw Data/20260716_LTREB_H2S_Plate2.csv"
##  [3] "Raw Data/20260716_LTREB_H2S_Plate3.csv"
##  [4] "Raw Data/20260716_LTREB_H2S_Plate4.csv"
##  [5] "Raw Data/20260716_LTREB_H2S_Plate5.csv"
##  [6] "Raw Data/20260717_LTREB_H2S_Plate10.csv"
##  [7] "Raw Data/20260717_LTREB_H2S_Plate6.csv"
##  [8] "Raw Data/20260717_LTREB_H2S_Plate7.csv"
##  [9] "Raw Data/20260717_LTREB_H2S_Plate8.csv"
## [10] "Raw Data/20260717_LTREB_H2S_Plate9.csv"
## [11] "Raw Data/20260720_LTREB_H2S_Plate11.csv"
```

```
# Define the file path for QAQC log file - NO Need to change just check year
log_path <- here("Porewater QAQC Logs", "GCReW_H2S_QAQC_Log.csv")

# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCReW_CO2xN_Porewater_H2S_202607.csv"
rerun_path <- "Processed Data/GCReW_CO2xN_Porewater_H2S_202607_reruns.csv"
```

##Read in metadata and create similar sample IDs for matching to samples

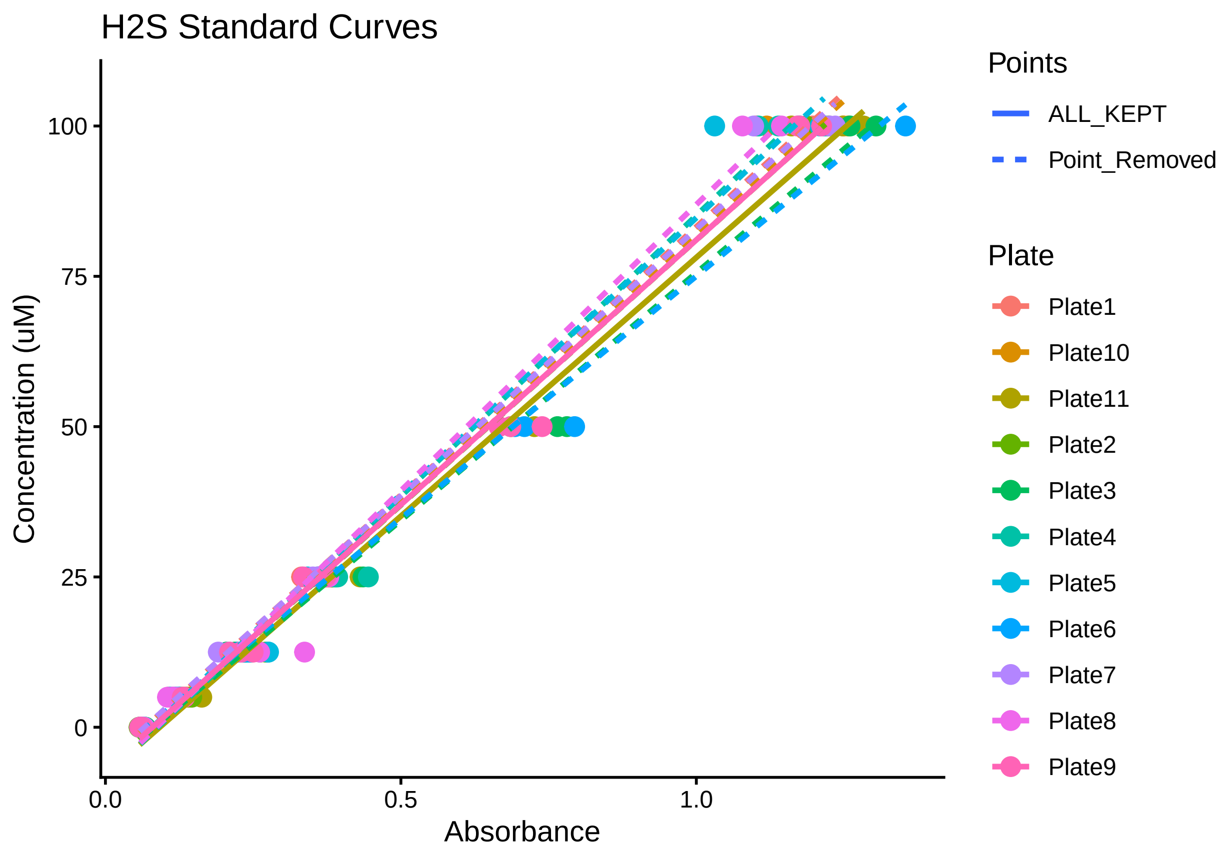
##Read in data

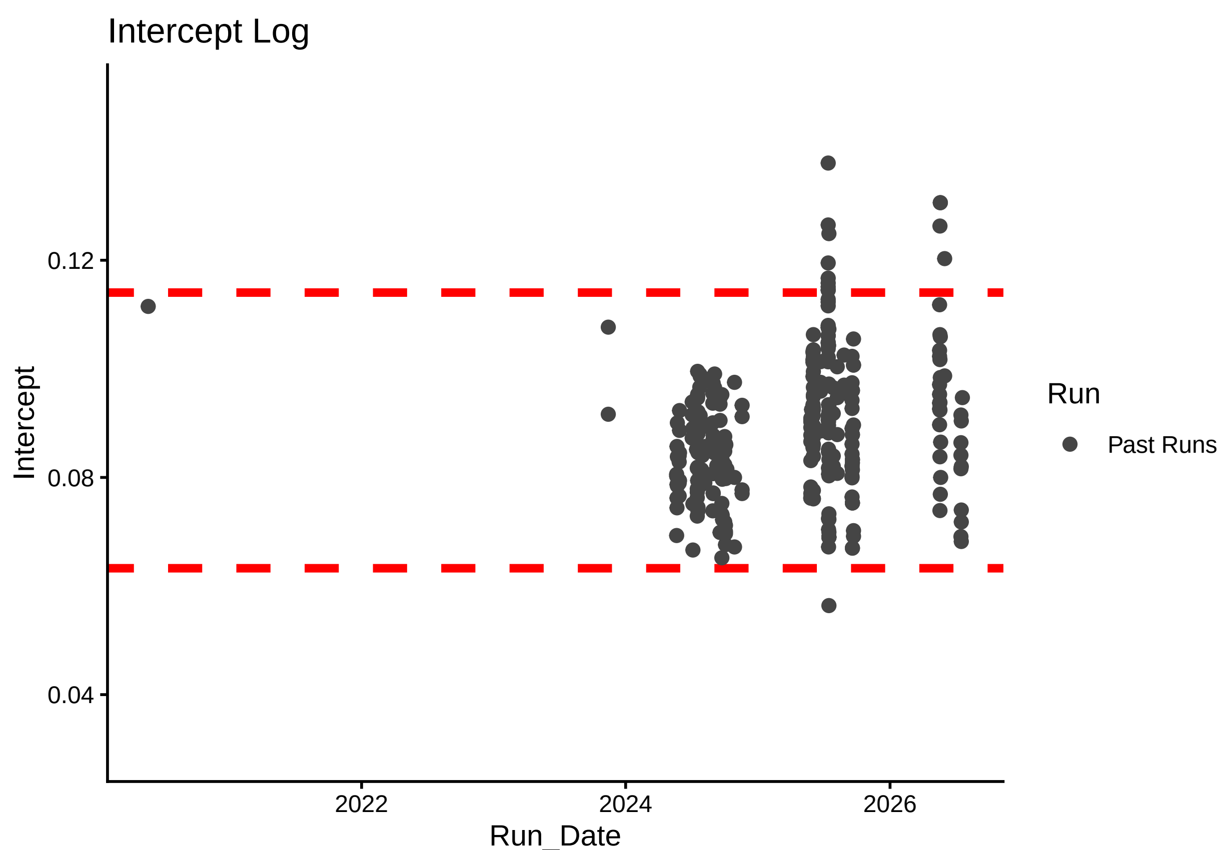
##Remove High CV Stds

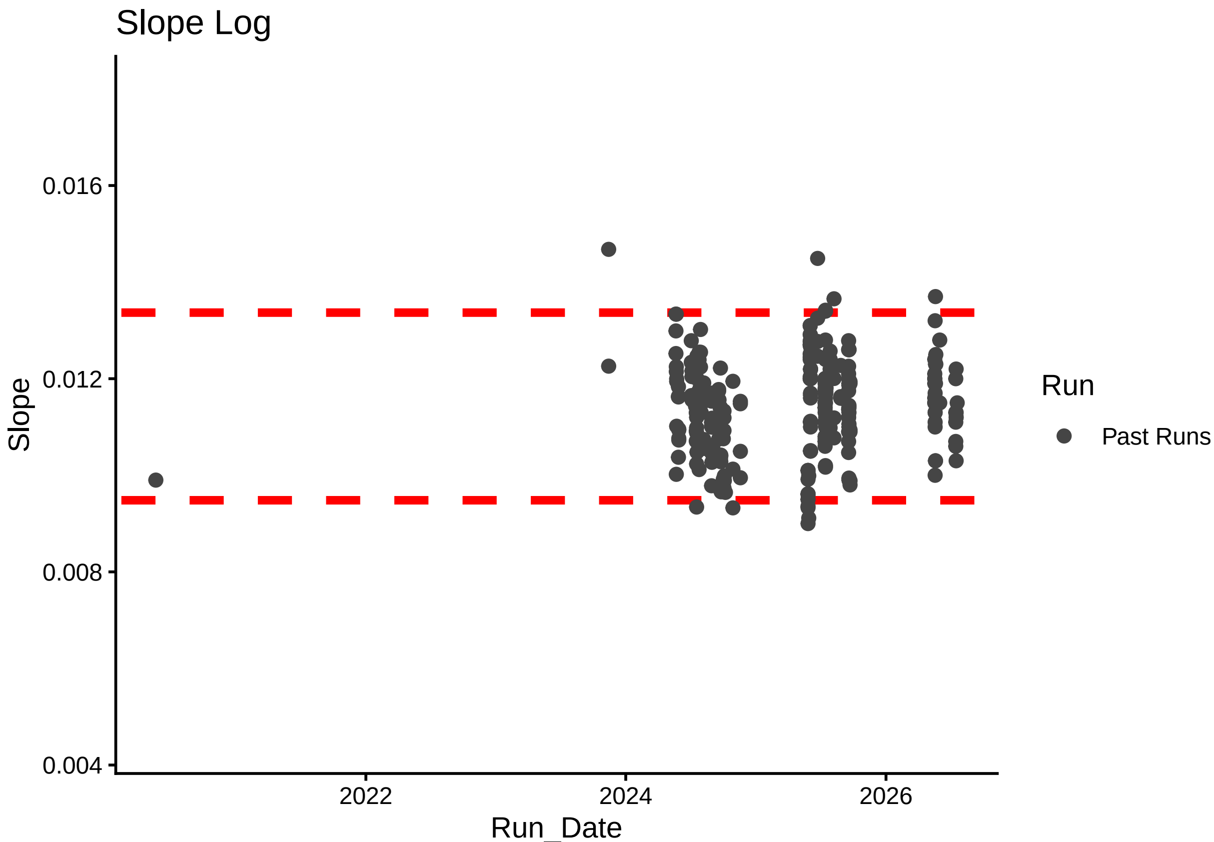
##Check R2 & Remove Standards if Necessary

```
## Joining with ‘by = join_by(Run_Date, Plate, IDs)‘
```

Plot Standard Curves







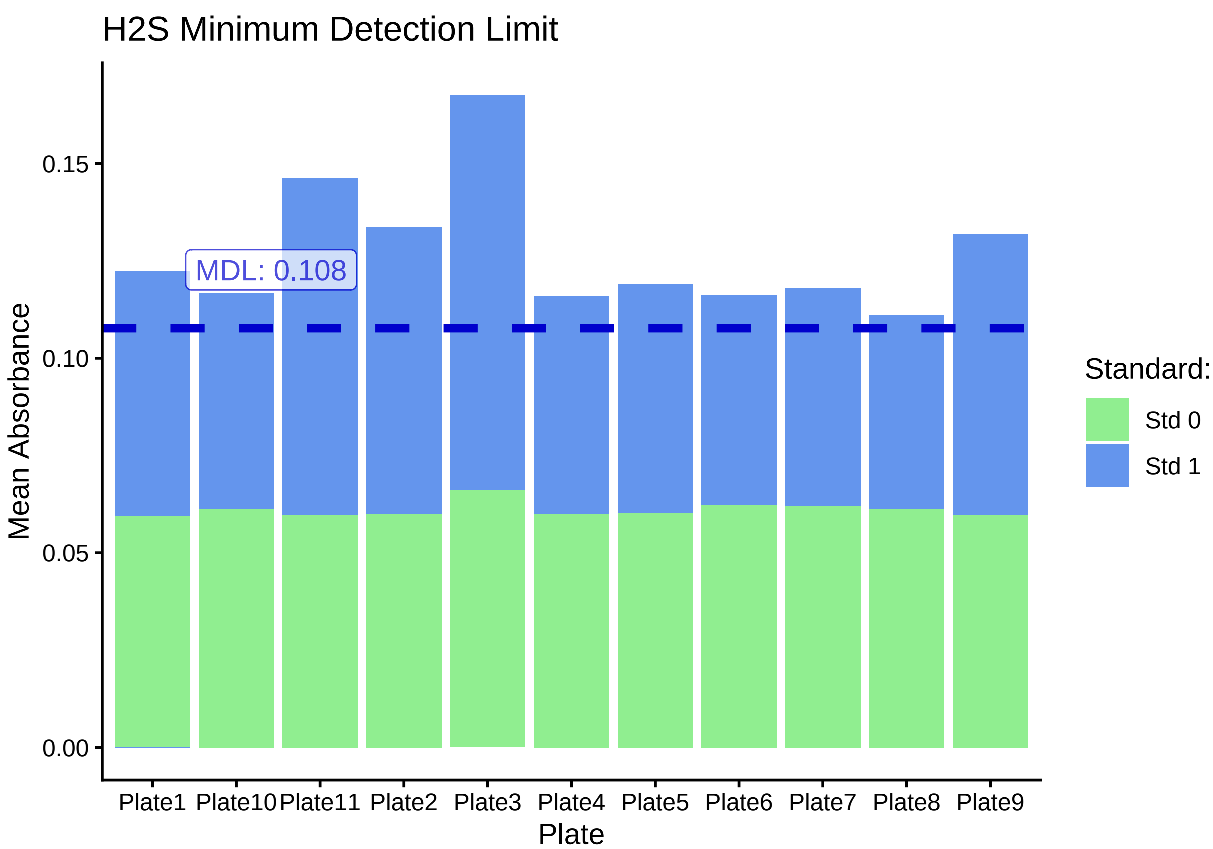
Return Flags For Standard Curve

Plate	R2_Flag	CV_Flag
Plate1	R2_Good	CV_Good
Plate10	R2_Good	CV_Good
Plate11	R2_Good	CV_Good
Plate2	R2_Good	CV_Good
Plate3	R2_Good	CV_Good
Plate4	R2_Good	CV_Good
Plate5	R2_Bad	CV_Good
Plate6	R2_Good	CV_Good
Plate6	R2_Good	CV_High
Plate7	R2_Good	CV_Good
Plate8	R2_Bad	CV_Good
Plate8	R2_Bad	CV_High
Plate9	R2_Good	CV_Good

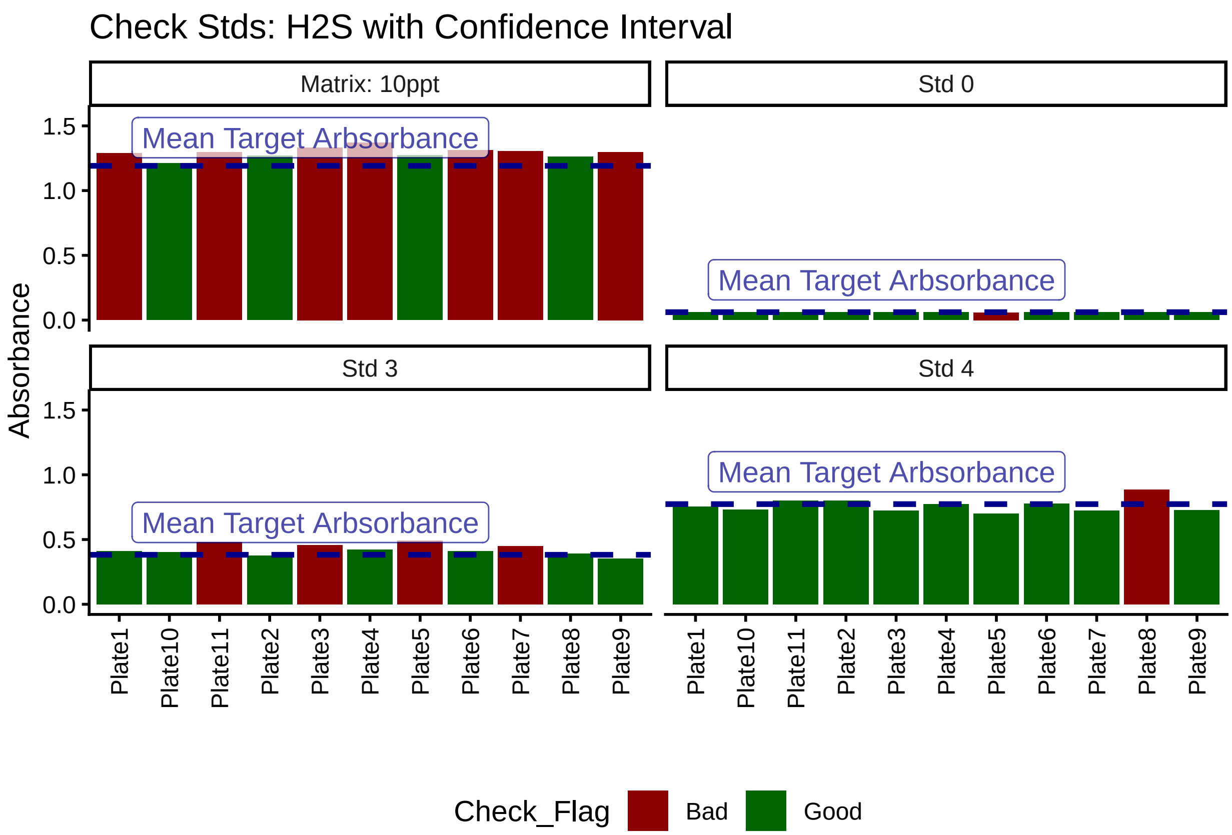
Plate	Curve_RMVD
Plate1	KEPT
Plate10	KEPT
Plate11	KEPT
Plate2	KEPT
Plate3	KEPT
Plate4	KEPT
Plate5	REMOVED
Plate6	KEPT
Plate7	KEPT

Plate	Curve_RMVD
Plate8	REMOVED
Plate9	KEPT

Method Detection Limit



Compare Check Standards to Standards



Display Any Check Flags

Plate	CHKs_Flag
Plate1	Std5Chk_Bad
Plate3	Std3Chk_Bad
Plate3	Std5Chk_Bad
Plate4	Std5Chk_Bad
Plate5	Std0Chk_Bad
Plate5	Std3Chk_Bad
Plate6	Std5Chk_Bad
Plate7	Std3Chk_Bad
Plate7	Std5Chk_Bad
Plate8	Std4Chk_Bad
Plate9	Std5Chk_Bad
Plate11	Std3Chk_Bad
Plate11	Std5Chk_Bad

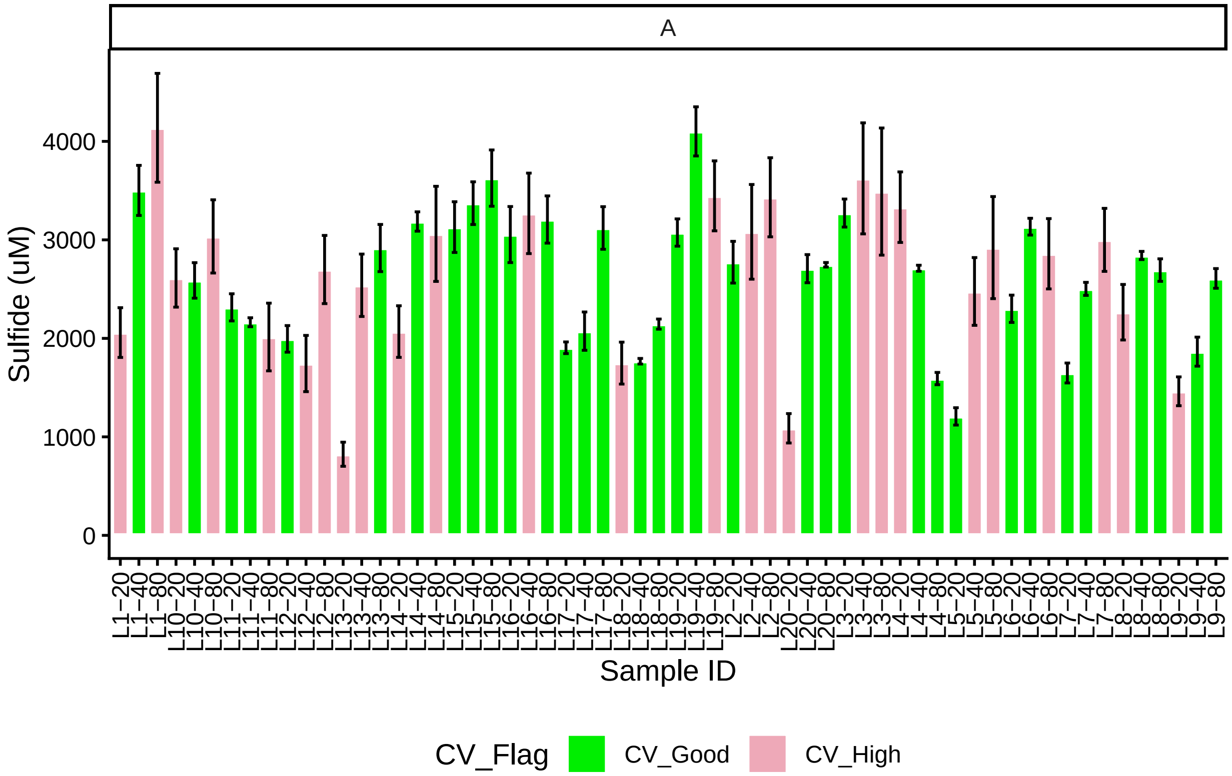
Plate	CHK_CV_Flags
Plate1	Std3Chk_CV_High
Plate1	Std4Chk_CV_High
Plate2	Std3Chk_CV_High
Plate3	Std3Chk_CV_High
Plate4	Std4Chk_CV_High

##Calculate Sulfide Concentrations & Add Concentration Flags

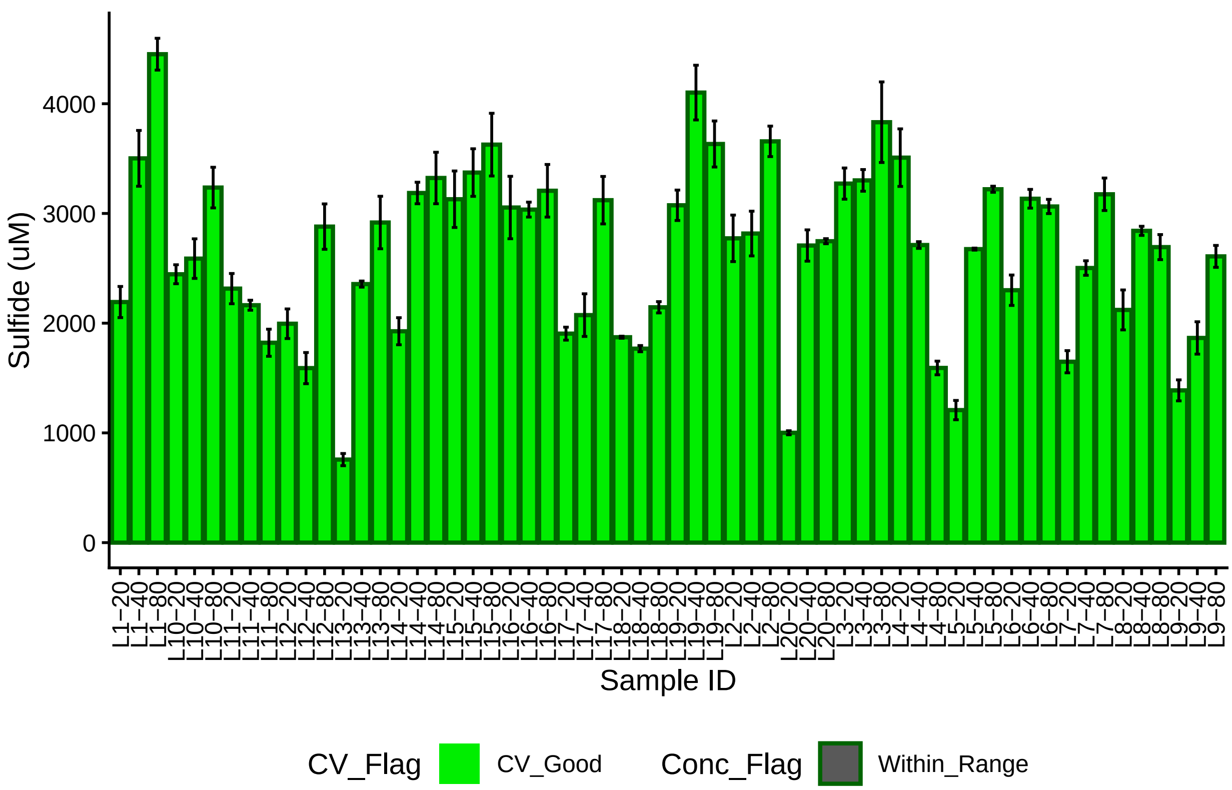
##Remove High CV Samples and Average

Plot Samples

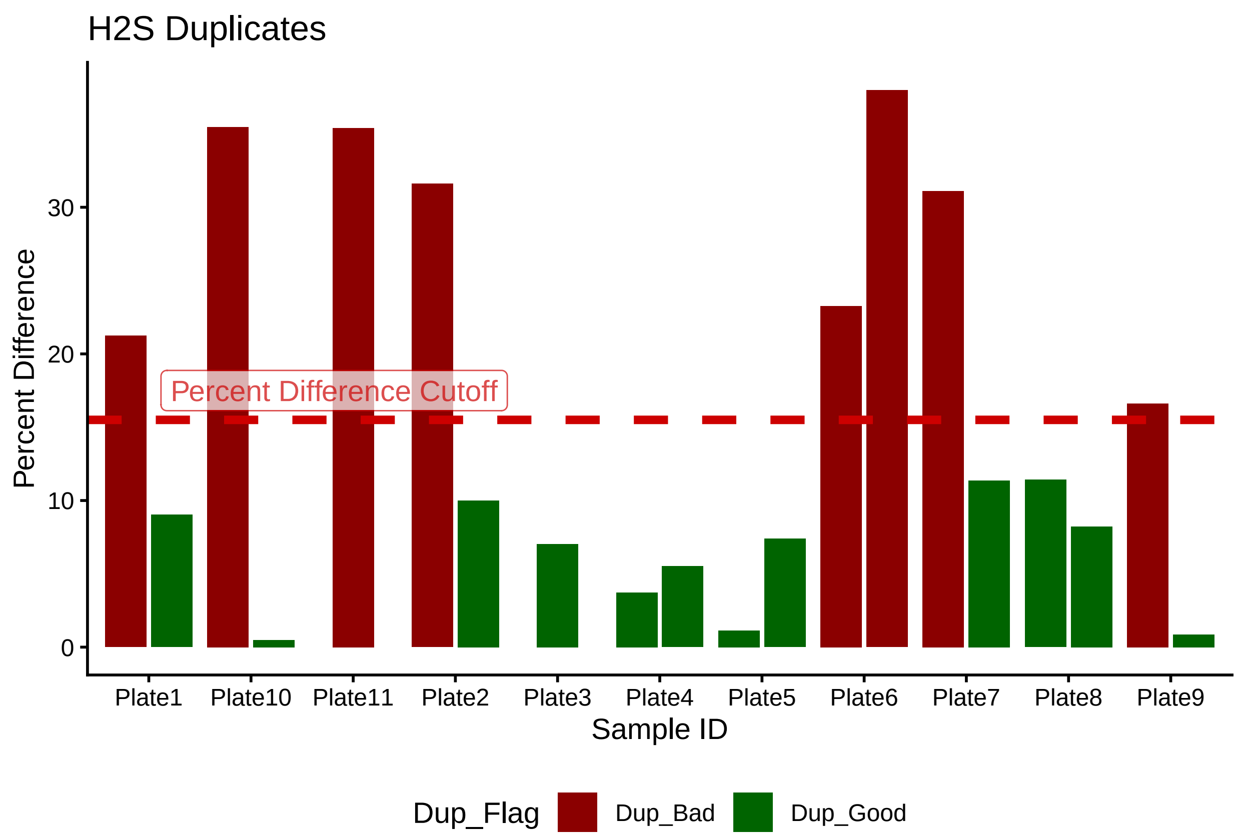
Sample triplicate means and sd dev before bad reps removed



Sample triplicate means and sd dev after bad reps removed



Duplicate QAQC



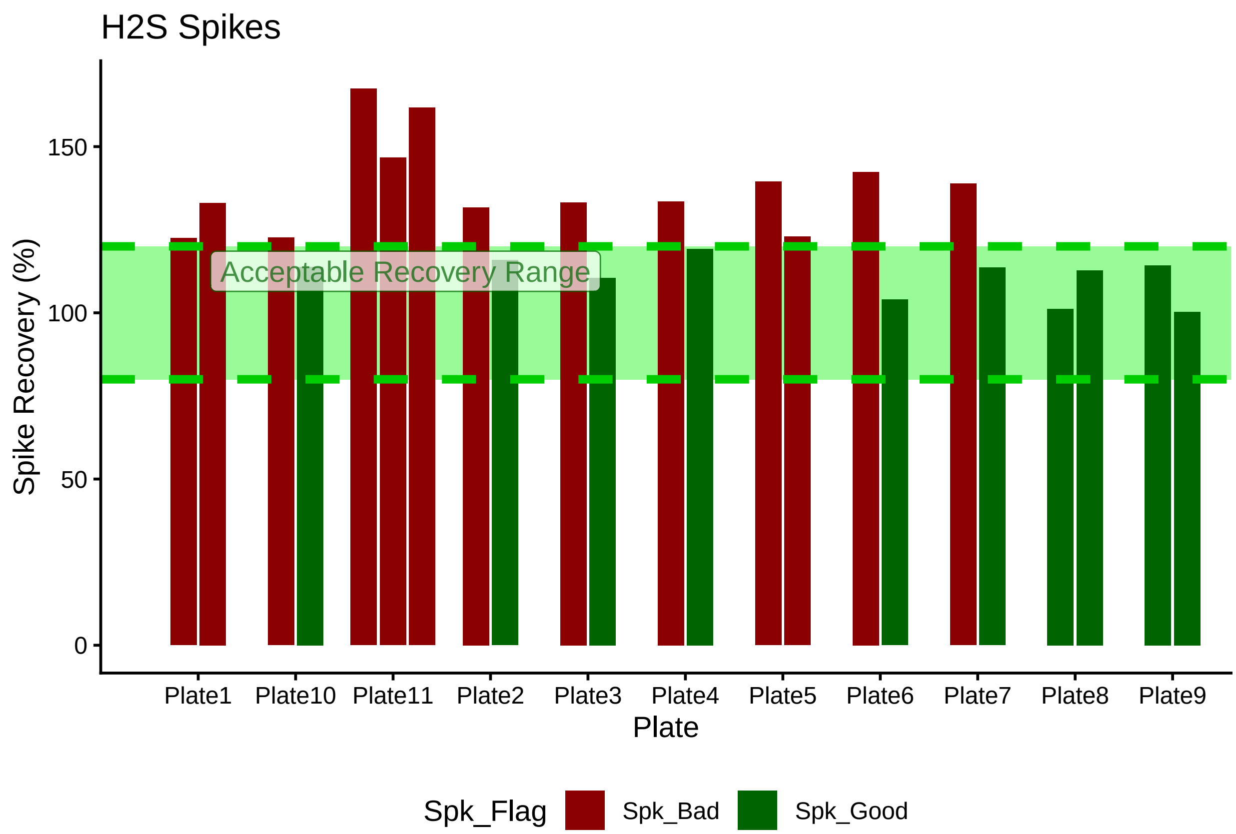
Display Dup Flags

```
## [1] ">60% Dups Pass"
```

Plate	Dup_Flag
Plate1	Dup_Bad
Plate2	Dup_Bad
Plate3	Dup_Good
Plate4	Dup_Good
Plate5	Dup_Good
Plate10	Dup_Bad
Plate6	Dup_Bad
Plate7	Dup_Bad
Plate8	Dup_Good
Plate9	Dup_Bad
Plate11	Dup_Bad

Spike QAQC

```
## Warning in left_join(., spk_flags_1, by = join_by(Spk_ID, Plate)): Detected an unexpected many-to-many r
## i Row 43 of `x` matches multiple rows in `y`.
## i Row 2 of `y` matches multiple rows in `x`.
## i If a many-to-many relationship is expected, set `relationship =
##   "many-to-many"` to silence this warning.
```



Display Spike Flags

```
## [1] "Spikes Paired"
```

Plate	Spike_Flag
Plate1	Spk_Bad
Plate2	Spk_Bad
Plate3	Spk_Bad
Plate4	Spk_Bad
Plate5	Spk_Bad
Plate10	Spk_Bad
Plate6	Spk_Bad
Plate7	Spk_Bad
Plate8	Spk_Good
Plate9	Spk_Good
Plate11	Spk_Bad

```
## [1] "<60% of Spikes Pass, Rerun"
```

Merge samples with Metadata & check all samples are present

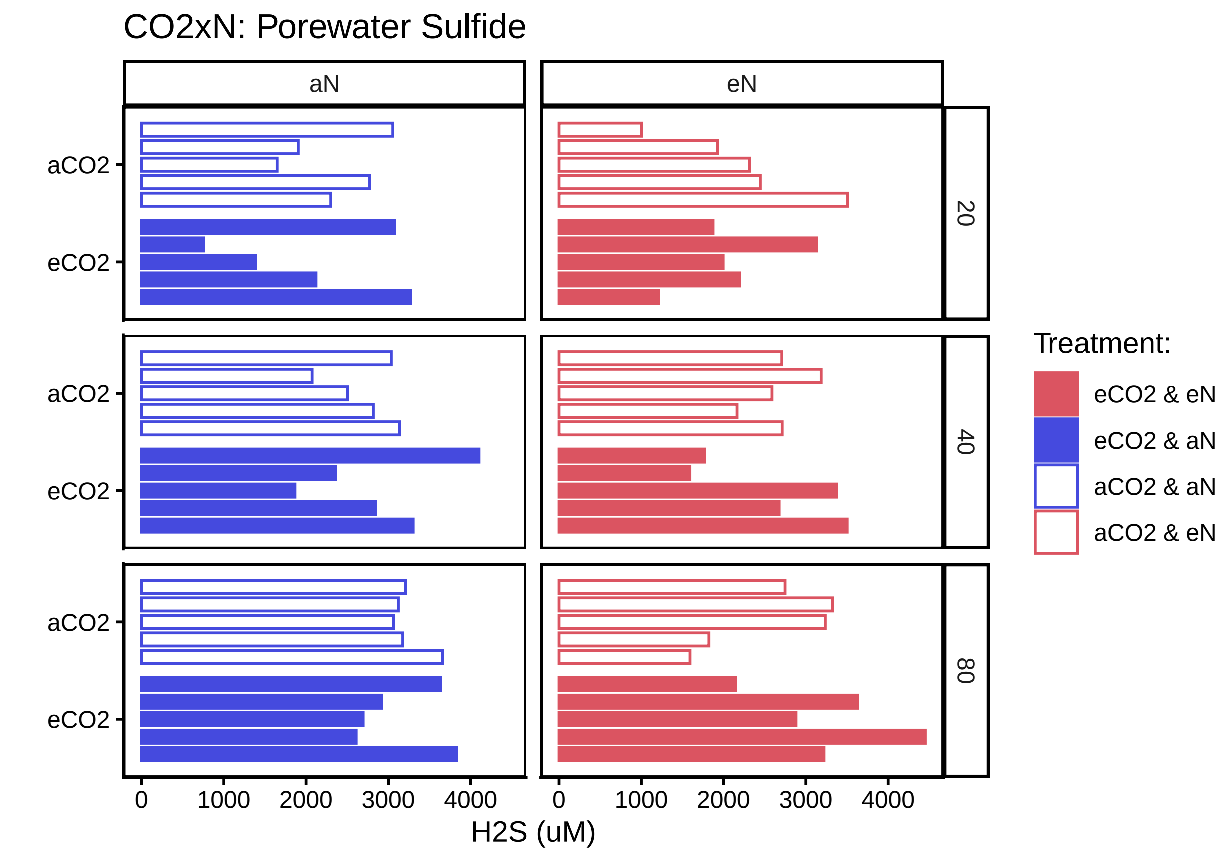
```
## All sample IDs are present in data
```

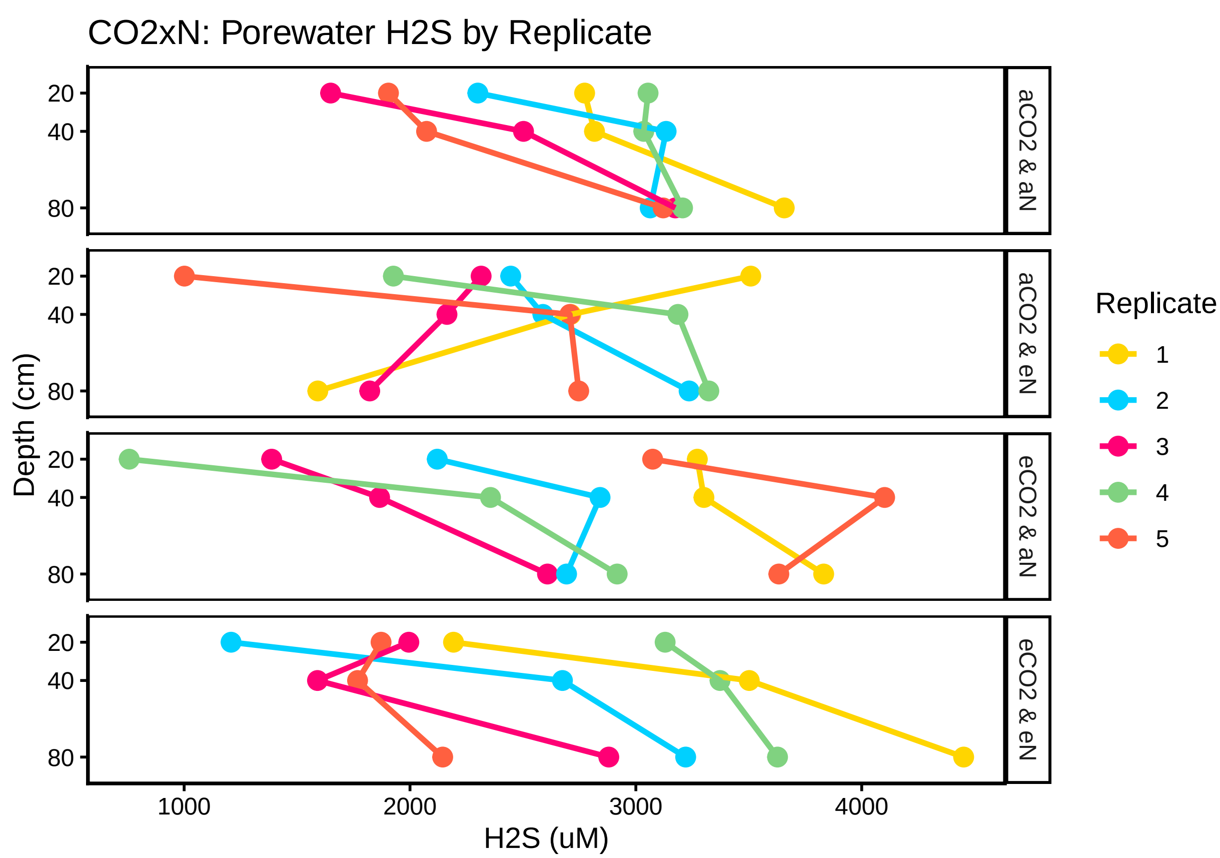
##Add Flags

##Organize Data

```
## Adding missing grouping variables: `Sample_ID.x`
## Adding missing grouping variables: `Sample_ID.x`
```

Visualize Data





##Export Data

END